

FSD(Full Stack Development)

HTML (Hypertext Markup Language): Structure / Content

Basic Structure of an HTML Document:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Web Page</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>This is my first HTML page.</p>
</body>
</html>
```

Explanation:

- **<!DOCTYPE html>**: Tells the browser that is an HTML5 document.
- **<html>**: The root element of the page.
- **<head>**: Contains metadata such as the page title, css links, etc.
- **<title>**: Set the title show on the browser tab.
- **<body>**: Contain the visible content of the web pages.
- **<h1> to <h6>** : Heading tag (h1 is the largest).
- **<p>**: Paragraph tag.

Common HTML Tag:

Tag	Purpose
<h1>--<h6>	Headings
<p>	Paragraph
<a>	Hyperlink
	Image
	Unordered list
	Ordered list
	List item
<table>	Table
<form>	Form
<input>	User input field
<button>	Button
<div>	Block-level container
	Inline container

HTML Attribute: HTML attribute provide additional information about html elements. They are written inside the opening tab.

Example:

```
<a href="https://www.google.com" target="_blank">Visit Google</a>
```

- **target="_blank"**: The target attribute specifies where to open the linked document.

When you use:the link opens in a new browser tab or window.

- **Href** is the attribute.
- **"https://www.google.com"** is the attribute value.

Other Target Value:

Value	Meaning
_self	Opens in the same tab (default)
_blank	Opens in a new tab/window
_parent	Opens in the parent frame
_top	Opens in the full browser window

Common HTML attributes:

Attribute	Used With	Purpose
href	<a>	Specifies the URL of a link
src	, <script>	Specifies the file path
alt		Alternative text for images
width	, <video>	Sets width
height	, <video>	Sets height
id	Any element	Unique identifier
class	Any element	Groups elements for CSS/JavaScript
style	Any element	Inline CSS styling
title	Any element	Tooltip text
value	<input>, <button>	Default value
placeholder	<input>	Hint text inside input
type	<input>, <button>	Specifies the input type
disabled	Form elements	Disables the element
required	Form elements	Makes a field mandatory

Parent and child Structure:

HTML element are organised in a tree-like Structure . An element that contains another element is called the **Parent**, and the element inside is called the **child**.

Example:

```
<div>
  <h1>Welcome</h1>
  <p>This is a paragraph.</p>
</div>
```

Structure:

```
div
├── h1
└── p
```

- Basic tag to create the Parent and Child Structure is **<div>** tag.

Term must Should Know:

Term	Meaning
Parent	Element that contains another element
Child	Element inside a parent element
Sibling	Elements with the same parent
Ancestor	Any element above another element
Descendant	Any element nested inside another element

Semantic and Non-Semantic:

Semantic: Semantic element clearly describe the meaning of the role of content they contain.

Example:

```
<header>Website Header</header>
<nav>Navigation Links</nav>
<main>Main Content</main>
<section>A Section of Content</section>
<article>Independent Article</article>
<aside>Sidebar Content</aside>
<footer>Website Footer</footer>
```

Example Code:

```
<body>
  <header>
    <h1>My Blog</h1>
  </header>

  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
  </nav>

  <main>
    <article>
      <h2>HTML Basics</h2>
      <p>Learning semantic HTML.</p>
    </article>
  </main>

  <footer>
    <p>Copyright 2026</p>
  </footer>
</body>
```

Non-Semantic:

Non-Semantic element do not describe the meaning of the there content . They are mainly used for grouping and styling.

Easy way to Remember:

Semantic: “What is the Content”

Non-Semantic: “ Just a Container”

Example:

```
<div>This is a container.</div>
<span>Inline container</span>
```

Example Code:

```
<div class="header">  
  <div class="logo">My Website</div>  
</div>
```

Block VS Inline element:

Block element: Block elements start on a **new line** and take up the **full width available** by default.

Characteristics:

- Start on a new line.
- Occupy the entire width of their parent container.
- Can contain both block and inline elements.

Example:

```
<div>This is a div.</div>  
<p>This is a paragraph.</p>  
<h1>Heading</h1>
```

Output:

This is a div.
This is a paragraph.
Heading
Section Content

Inline element:

Inline elements **do not start on a new line**. They only take up as much width as needed.

Character:

- Stay in the same line.
- Occupy only the required width.
- Generally should not contain block elements.

Example:

```
<span>Hello</span>  
<a href="#">Link</a>  
<strong>Bold Text</strong>  
<section>Section Content</section>
```

Output:

Hello Link Bold Text

Comparison:

Feature	Block Elements	Inline Elements
Starts on a new line?	Yes	No
Takes full width?	Yes	No
Width/Height can be set?	Yes	Limited
Can contain block elements?	Yes	Usually No
Examples	<div>, <p>, <h1>	, <a>,

How to attach image in HTML:

To display an image in HTML, use the ** tag**.

Syntax: ``

Attributes:

- **src** → Specifies the path (location) of the image.
- **alt** → Alternative text shown if the image cannot be displayed.

If image are having in different(Parent Folder) folder:

Suppose:

```
project/  
├── photo.jpg  
└── pages/  
    └── index.html
```

From pages/index.html:

``

Explanation:

- `..` → Go up from pages to project
- `images/` → Enter the images folder
- `photo.jpg` → Load the image

HTML LISTs:

HTML Lists are used to display a group of related item. There are Three Types of List in HTML.

- **Ordered List** : An Ordered list item are display in specific order using number or letter.

Syntax:

```
<ol>  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ol>
```

Output:

1. Item 1
2. Item 2
3. Item 3

- **Unordered List**: An Unordered List displays item using bullets.

Syntax:

```
<ul>  
  <li>Apple</li>  
  <li>Banana</li>  
  <li>Mango</li>  
</ul>
```

Output:

- Apple
- Banana
- Mango
- **Description List**: Used to display there term and there description:

`<dl>` → Description List

`<dt>` → Description Term

<dd> → Description Details

Example:

```
<dl>
  <dt>HTML</dt>
  <dd>A markup language for web pages.</dd>

  <dt>CSS</dt>
  <dd>Used for styling web pages.</dd>
</dl>
```

Output:

HTML
A markup language for web pages.
CSS
Used for Styling the pages

Notes:

The **<hr>** tags in HTML are stand for Horizontal rule.

The **
** tags in HTML are used for moving in next line.

HMTL TABLES:

HTML tables are used to display data in **row** or **columns**

Code Example:

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>Ritanshu</td>
    <td>21</td>
    <td>Arrah</td>
  </tr>
  <tr>
    <td>Aman</td>
    <td>22</td>
    <td>Patna</td>
  </tr>
</table>
```

Output:

Name	Age	City
------	-----	------

Ritanshu	21	Arrah
Aman	22	Patna

Important Table Tags:

Tag	Meaning
<table>	Creates a table
<tr>	Table Row
<th>	Table Header Cell
<td>	Table Data Cell
<caption>	Adds a title to the table
<thead>	Groups header rows
<tbody>	Groups body rows
<tfoot>	Groups footer rows

Tables are having a three children:

1. Table Head
2. Table Body
3. Table Foot

Column-span Attributes:

Used to make cell span multiple columns

Code:

```
<table border="1">
  <tr>
    <th colspan="2">Student Information</th>
  </tr>
  <tr>
    <td>Name</td>
    <td>Ritanshu</td>
  </tr>
</table>
```

HTML Forms:

HTML form are used to take the input from the user like name, email, password, etc..

Basic form structure:

Code:

```
<form>
  <label>Name:</label>
  <input type="text">
  <br><br>
  <label>Email:</label>
  <input type="email">
  <br><br>
  <input type="submit" value="Submit">
</form>
```

Output:

Name: _____
Email: _____
[Submit]

Why we Use label tags:

The **<label>** tag is used to **describe an input field** in a form. It tells users what information they need to enter

Example without <labels>:

Name:

```
<input type="text">
```

This works, but the text "Name:" is not associated with the input field.

Example with< Labels>

```
<label for="name">Name:</label>
<input type="text" id="name">
```

<Form> Tag:

The form element that contains all form controls.

Code:

```
<form action="/submit" method="post">
  ...
</form>
```

Attributes for forms

Attribute	Description
action	URL where form data is sent
method	HTTP method (get or post)
autocomplete	Enables/disables auto-fill
target	Specifies where to display the response

Input Tags:

Used to create different type of input fields

```
<input type="text">
```

Common Input Types:

Type	Purpose
text	Single-line text
password	Password field
email	Email input
number	Numeric input
date	Date picker
radio	Single selection
checkbox	Multiple selections
file	File upload
submit	Submit button
reset	Reset form
button	Normal button

TextInput:

<label>Username:</label>

<input type="text" name="username">

Password Input:

<label>Password:</label>

<input type="password" name="password">

Email Input:

<label>Email:</label>

<input type="email" name="email">

Min Length and Max Length in HTML

They are used to control the length if the input character.

Max-length: Maximum number of character allowed.

Min-length: Minimum number of character allowed.

Task: Create a form using HTML.